

Multiple Choice

1. **Answer:** B

Options: A) $P(E, F), P(H), P(E|H), P(F|H)$; B) $P(E, F), P(H), P(E, F|H)$; C) $P(H), P(E|H), P(F|H)$; D) $P(E, F), P(E|H), P(F|H)$

Note: Correct option: B - $P(E, F), P(H), P(E, F|H)$

2. **Answer:** A

Options: A) Linear hard-margin SVM.; B) Linear Logistic Regression.; C) Linear Soft margin SVM.; D) The centroid method.

Note: Correct option: A - Linear hard-margin SVM.

3. **Answer:** A

Options: A) Supervised learning; B) Unsupervised learning; C) Clustering; D) None of the above

Note: Correct option: A - Supervised learning

4. **Answer:** A

Options: A) linear in K ; B) quadratic in K ; C) cubic in K ; D) exponential in K

Note: Correct option: A - linear in K

5. **Answer:** D

Options: A) It relates inputs to outputs.; B) It is used for prediction.; C) It may be used for interpretation.; D) It discovers causal relationships

Note: Correct option: D - It discovers causal relationships

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $O(N)$ '.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The use of weak classifiers'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '8'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** `def swap_count(s): | return swap`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def nth_items(list,n): | return list[:n]`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key